

(Please write your Exam Roll No.)

Exam Roll No.....

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] Jan-Feb 2023

Paper code: ECC-209

Subject: Analog Communication

Time: 3:00 hours

Maximum Marks: 75

Note: Attempt five questions in all including Q. no.1 which is compulsory. Select one question from each unit.

Q 1. Answer the following

(1 X 15 = 15)

- (a) Write an expression for DSB_SC modulated wave.
- (b) What is the bandwidth of SSB Signal?
- (c) If A_c is 20 and the modulation index is 0.5. Determine the value of A_m .
- (d) What is meant by the Fourier spectrum?
- (e) What is Carson's Rule?
- (f) How much is the maximum frequency deviation in FM?
- (g) Define deviation ratio in FM.
- (h) Why is the local oscillator frequency always higher than the signal frequency?
- (i) What is a random process?
- (j) What do you mean by PDF?
- (k) What is the ergodic process?
- (l) Define power spectral density.
- (m) Define FOM in a receiver system.
- (n) Define SNR.
- (o) What do you mean by superposition of noise?

UNIT-I

Q 2: (a) Draw the block diagram of a communication system and explain the function of each block. (8)

(b) Define the Frequency shifting property of the Fourier Transform. Find the Fourier transform of $\cos(at)$. (7)

Q 3: (a) A sinusoidal carrier has an amplitude of 10 V and a frequency of 30 KHz. It is amplitude modulated by a sinusoidal signal having an amplitude of 3 V and frequency of 1 KHz. (i) write the equation of modulated wave. (ii) Plot the modulated wave showing maxima and minima in the waveform. (iii) Determine modulation index. (iv) draw spectrum. (8)

(b) Explain the envelope detector circuit with the importance of the RC time constant. (7)

UNIT-II

Q 4 (a) Derive the expression for single tone frequency modulated wave. (8)

(b) A FM wave is represented by $v(t)$. Find the Carrier frequency and modulating frequency with modulation index and maximum deviation.

$$V(t) = 10\sin(5 \times 10^8 t + 4\sin(1250t)) \quad (7)$$

Q 5 (a) Explain the PLL method of FM detection. (8)

(b)) Explain the Armstrong method for wideband FM generation. (7)

UNIT- III

Q 6 (a) What are Probability and cumulative distribution function (CDF)? Explain their important properties.. (8)

(b) A certain random variable is given by

$$F_X(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ kx^2 & \text{for } 0 < x \leq 10 \\ 100k & \text{for } x \geq 10 \end{cases}$$

(i) Find k ii) Find $P(x \leq 5)$ and iii) Find $P(5 < x \leq 7)$ (7)

Q 7 (a) Explain binomial distribution and find its mean and variance. (8)

(b) Explain the different classification of the random process. (7)

UNIT-IV

Q 8 (a) Explain the receiver model with a diagram and comment on its utility. (8)

(b) Explain the band-pass noise model. (7)

Q 9 (a) Derive the expression for FOM of the DSB_SC receiver. (10)

(b) Explain the quadrature components of noise. (5)